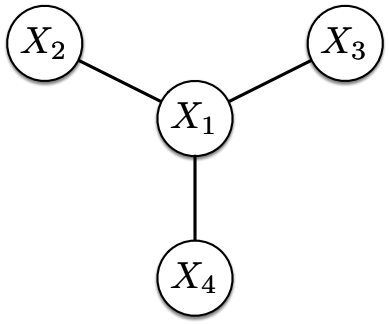
Massachusetts Institute of Technology
Department of Electrical Engineering and Computer Science
6.3800 Introduction to Inference
Fall 2026

Recitation 9 Handout

Thursday, October 8

Problem 9.1

Let $X_1, X_2, X_3, X_4 \in \mathcal{X} = \{0, 1\}$ be binary random variables and consider the following tree graphical model representing the structure in the joint distribution p_{X_1,X_2,X_3,X_4}:



Moreover, suppose

$$\phi_{1,2}(x, x') = \phi_{1,3}(x, x') = \phi_{1,4}(x, x') = \begin{cases} 3 & x = 0,\ x' = 1 \\ 1 & \text{otherwise,} \end{cases}$$

$$\phi_2(x) = \phi_3(x) = \phi_4(x) = 1, \qquad x \in \mathcal{X},$$

$$\phi_1(x) = \begin{cases} \alpha & x = 1 \\ 1 & x = 0 \end{cases}$$

where $\alpha > 0$ is a constant.

(a) Compute all sum-product algorithm messages.

(b) Compute all marginal distributions.

(c) Determine the range of values α for which $p_{X_i}(1) > p_{X_i}(0)$, i.e., X_i is more likely to be 1 than 0, $i = 1, 2, 3, 4$.